Azure status history

This page contains Post Incident Reviews (PIRs) of previous service issues, each retained for 5 years. From November 20, 2019, this included PIRs for all issues about which we communicated publicly. From June 1, 2022, this includes PIRs for broad issues as described in our documentation.

Product:

All

Region:

All

Date:

All

July 2024

Post Incident Review (PIR) – Managed Identities for Azure resources – Failures in Australia East

Tracking ID: FTVR-L9Z

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/FTVR-L9Z

What happened?

A platform issue caused some of our 'Managed Identities for Azure resources' (formerly Managed Service Identity or MSI) customers to experience failures when requesting managed identity tokens or performing management operations for managed identities in the Australia East region, between 00:55 UTC and 06:28 UTC on 12 July 2024. All identities in this region were subject to impact.

For managed identities for Azure Virtual Machines (VMs) and Virtual Machine Scale Sets (VMSS) in this region, around 3% of token requests failed during this time. Established virtual machines with steady token activity in the days and hours prior to the incident were generally not impacted. For management operations, around 50% of calls via Azure Resource Manager (ARM) or the Azure Portal to perform operations such as creating, deleting, assigning, or updating identities failed during this time.

The creation of any Azure resources to which system-assigned managed identities were assigned would also have failed, since the creation of these resources requires the creation of a managed identity. Identity management and managed identity token issuance for other Azure resource types - including Azure Service Fabric, Azure Stack HCI, Azure Databricks, and Azure Kubernetes Service - were impacted to varying degrees depending on the specific scenario and resource. Generally, across all Azure compute resource types, management operations were heavily impacted. For token requests, existing resources continued to function during the incident, while new or newly scaled resources saw failures.

What went wrong and why?

At Microsoft, we are working to remove manual key management and rotation from the operation of our internal services. For most of our infrastructure and for our customers, we recommend migrating authentication between Azure resources to use Managed identities for Azure resources. For the Managed identities for Azure resources infrastructure itself, though, we cannot use Managed identities for Azure resources, due to a risk of circular dependency. Therefore, Managed identities for Azure resources uses key-based access to authenticate to its downstream Azure dependencies that store identity metadata.

To improve resilience, we have been moving Managed identities for Azure resources infrastructure to an internal system that provides automatic key rotation and management. The process for rolling out this change was to deploy a service change to support automatic keys, and then roll over our keys into the new system in each region one-by-one, following our safe deployment process. This process was tested and successfully applied in several prior regions.

In the Australia East and Australia Southeast regions, however, the required deployment for key automation did not complete successfully - because these regions had previously been pinned to an older build of the service, for unrelated reasons. Our internal deployment tooling reported a successful deployment and did not clearly show that these regions were still pinned to the older version. Believing the

deployment to have completed, a service engineer initiated the switch to move to automatic keys for the Australia East storage infrastructure, immediately causing the key to roll over. Because neither Australia East nor its failover pair Australia Southeast were running the correct service version, they continued to try to use the old key when accessing storage for Australia East identities, which failed.

How did we respond?

12 This incident was detected within three minutes by internal monitoring of both the Managed Identity service, and other dependent services in Azure. Our engineers immediately knew that the cause was related to the key rollover operation that had just been performed. Since managed identities are built across Availability Zones and failover pairs, our standard mitigation playbook is to perform a failover to a different zone or region. However, in this case, all zones in Australia East were impacted - and requests to the failover pair, Australia Southeast, were also failing.

We pursued several investigation and mitigation threads in parallel. Our engineers initially believed that the issue was that the key rollover had somehow failed or applied incorrectly, and we attempted several mitigation steps to force the new key to be picked up or force the downstream resources to accept the new key. After these were not effective, we discovered that the issue was not with the new key, but rather that our instances were still attempting to use the old key - and that this key had been invalidated by the rollover. We launched more parallel workstreams to understand why this would be the case and how to mitigate, since we still believed that the deployment had succeeded. Once we realized that these regions were in fact pinned to an older build of the service, we authored and deployed a configuration change to force this build to use the managed key, which resolved the issue. Full mitigation occurred at 06:28 UTC on 12 July 2024.

How are we making incidents like this less likely or less impactful?

- We temporarily froze all changes to the Managed identities for Azure resources. (Completed)
- We ensured that all Azure regions with Managed Identity that could be subject to automatically managed keys are running a service version and configuration that supports those keys. (Completed)
- We are working to improve our deployment process to ensure that pinned build versions are detected as part of a deployment and not considered successful. (Estimated completion: July 2024)
- Prior to rolling out automatic key management again, we will improve our service telemetry to enable us to quickly see whether a manually or automatically managed key is used, as well as the identifier of the key. (Estimated completion: July 2024)
- Prior to rolling out automatic key management again, we will update our internal process to verify that all service instances are running the correct version and using the expected key. (Estimated completion: July 2024)
- For the future move to automatic key management, we will schedule the changes to occur in off-peak business hours in each Azure region. (Estimated completion: July 2024)
- Finally, we are working to improve the central documentation for automatic key management to ensure the above best practices are followed by all Azure services using this capability. (Estimated completion: July 2024)

How can customers make incidents like this less impactful?

- In Azure, our safe deployment practices consider both availability zones and Azure regions. We encourage all customers to partition their capacity across regions and zones, and to build and exercise a disaster recovery strategy to defend against issues that may occur in a single region. During this incident, customers who were able to fail out of Australia East would have been able to mitigate their impact. To learn more about a multi-region geodiversity strategy for mission-critical workloads, see https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application and https://learn.microsoft.com/azure/architecture/patterns/geodes. Finally, this page provides more information on disaster recovery strategies: https://learn.microsoft.com/azure/well-architected/reliability/disaster-recovery
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/en-us/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/FTVR-L9Z

April 2024

Post Incident Review (PIR) – DNS resolution issues impacted Azure in China Tracking ID: V_BH-7DZ

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/V_BH-7DZ

What happened?

Between 03:45 CST and 06:00 CST on 23 Apr 2024, a configuration change performed through a domain registrar resulted in a service disruption to 2 system domains (chinacloudapp.cn and chinacloudsites.cn) that are critical in our cloud operations in China. This caused the following impact in the Azure China regions:

Customers may have had issues connecting to multiple Azure services including Cosmos DB, Azure Virtual Desktop, Azure Databricks, Backup, Site Recovery, Azure IoT Hub, Service Bus, Logic Apps, Data Factory, Azure Kubernetes Services, Azure Policy, Azure AI Speech, Azure Machine Learning, API Management, Azure Container Registry, and Azure Data Explorer.

Customers may have had issues viewing or managing resources via the Azure Portal (portal.azure.cn) or via APIs.

Azure Monitor offerings, Log Analytics and Microsoft Sentinel in the China East 2 and China East 3 experienced intermittent data latency, failure to query and retrieve data which could have resulted in a failure of alert activation, and/or failures to create, update, retrieve or delete operations in Log Analytics.

What went wrong and why?

To comply with a regulatory requirement of the Chinese government, we conducted an internal audit, ensuring all our domains had the appropriate ownership and documented properly. During this process, ownership for two critical system domains for Azure in China were misattributed, and as a result, were flagged as potential candidates for decommission.

The next step of the decommissioning process is a period of monitoring active traffic on a flagged domain before continuing with its decommission. However, the management tool that provides DNS zone and hosting information was not scoped to include zones hosted within Azure China, which caused our system to report that the zone file did not exist. It is common for end-of-life domains that are not in use to not have a zone file, and, as such, the non-existent zone file notice did not raise any alerts with the operator. The workflow then proceeded to the next stage, where the nameservers of these two domains were updated to a set of inactive servers, which is a final check to identify any hidden users or services dependent on the domain.

As DNS caches across the Internet gradually timed out, DNS resolvers made requests to refresh the information for the two domains and received responses containing the inactive nameservers, resulting in failures to resolve FQDNs in those domains. Our health signals detected this degradation in our Azure China Cloud and alerted our engineers. Once we understood the issue, the change was reverted in a matter of minutes. However, the mitigation time was prolonged due to the caching applied by DNS resolvers.

The issue impacted only specific Microsoft-owned domains, and it did not affect the Azure DNS platform availability or DNS services serving any other zone hosted on Azure.

How did we respond?

- 03:45 CST on 23 April 2024 – Nameserver configuration was updated. Due to previous DNS TTLs (Time To Live) the impact was not immediate.
- 04:37 CST on 23 April 2024 - Our internal monitors alerted us to degradation in the service and created Incident.
- 04:39 CST on 23 April 2024 – Incident was acknowledged by our engineering team.
- 04:57 CST on 23 April 2024 – We determined the cause of the resolution failures coincide with a change in name servers for the chinacloudapp.cn and chinacloudsites.cn domains.
- 04:59 CST on 23 April 2024 - We reverted to use the previously known good name servers.
- 05:13 CST on 23 April 2024 - The reversion was completed, at which point services began to recover.
- 06:00 CST on 23 April 2024 - Full recovery was declared after verifying that traffic for the services and affected DNS zones had recovered back to pre-incident levels.

How are we making incidents like this less likely or less impactful?

- We have suspended any further runs of this domain lifecycle process until updates to the management tool for Azure in China are completed.
- Update our validation process regarding domain lifecycle management to ensure all cloud region signals are incorporated (Estimated completion: June 2024).
- Implement additional validations to obtain nameserver information by directly resolving the zone over internet (Estimated completion: July 2024).

How can our customers and partners make incidents like this less impactful?

- As this issue impacted two domains used in operating the management plane of the Azure China Cloud and naming Azure services offered in the Azure China Cloud, users of the China Cloud did not have many opportunities to design their services to be resilient to this type of outage.
- More generally customers should consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts: https://aka.ms/ash-alerts . These can trigger emails, SMS, webhooks, push notifications (via the Azure Mobile app https://aka.ms/AzureMobileApp) and more.

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/V_BH-7DZ

Post Incident Review (PIR) - Azure Database for MySQL/MariaDB in West US, Azure Databricks in 3 regions

Tracking ID: QS3K-BPZ

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/QS3K-BPZ

What happened?

Between 04:26 UTC and 07:30 UTC on 19 April 2024, a platform issue in the West US region resulted in impact for the following services:

- **Azure Database for MariaDB**: Connectivity issues, all connections may have failed for a subset of customers. Retries would have been unsuccessful.
- **Azure Database for MySQL - Single Server**: Connectivity issues, all connections may have failed for a subset of customers. Retries would have been unsuccessful.
- **Azure Databricks**: Due to Azure Databricks dependencies on the impacted database service, customers in West US, West US 2 and/or South Central US may have experienced failures and timeouts with workspace login and authentication requests. Cluster CRUD requests and jobs relying on cluster start/resize/termination may not have executed, jobs submitted through APIs/Schedulers may not have executed, UI and Databricks SQL queries may have timed out, users may have experienced failures launching Databricks SQL Serverless Warehouses and may have been unable to access UC APIs, and customers may have observed errors citing “Authentication is temporarily unavailable” or “TEMPORARILY_UNAVAILABLE”.

What went wrong and why?

All connections to Azure Database for MySQL - Single Server and Azure Database for MariaDB service are established through a gateway responsible for routing incoming connections to their respective servers. This gateway service is hosted by a group of stateless compute nodes sitting behind an IP address. As part of ongoing service maintenance, compute hardware hosting the gateway service is periodically refreshed to ensure we provide the most secure and performant experience. Before moving the incoming new connections requests to a new gateway ring, customers running their servers and connecting to older gateway rings are notified via email and in the Azure portal to update their outbound rules to allow new gateway IP address.

When the gateway hardware is refreshed, a new ring of the gateway compute nodes is built out first. This new ring serves the traffic for all the newly created Azure Database for MySQL servers, and it will have a different IP address from older gateway rings in the same region, to differentiate the traffic. Once the new ring is fully functional, existing server connection traffic is routed to the new gateway by updating the DNS - and the older gateway hardware serving existing servers is planned for decommissioning. After the DNS update, all newer connections from the clients are automatically routed to the new gateway ring.

On 19 April 2024, as part of the planned maintenance event, the move of existing server incoming traffic to the new gateway ring was scheduled in West US region. At 04:26 UTC, the DNS update was made for a batch of Azure Database for MySQL and MariaDB servers, to route new incoming connections for servers to a new gateway ring. Due to a procedural error, the new connections were erroneously routed to a newly built gateway ring whose configuration was incomplete. This new gateway ring was not ready to accept new connections yet - this led to the login requests failing at the gateway, and therefore the gateway rejecting all new incoming connections to the Azure Database for MySQL and Azure Database for MariaDB servers. We initially failed to detect the failures caused by this change, as the new gateway ring configuration was incomplete, so it did not have the telemetry and monitoring configured yet. As a result, the issue went undetected from service telemetry, until we received reports from internal teams about unreachable databases.

How did we respond?

- 19 April 2024 @ 04:46 UTC - Our MySQL team received a report from Azure Databricks that multiple databases were unreachable.
- 19 April 2024 @ 05:14 UTC - Our MySQL on-call engineers determined that the databases were healthy but that no connections were being made.
- 19 April 2024 @ 05:44 UTC - Our networking team was engaged to assist with the investigation.
- 19 April 2024 @ 06:46 UTC - The issue was correlated to the attempt to move traffic to the new gateway ring, and mitigation steps to roll back the change were initiated.
- 19 April 2024 @ 07:30 UTC - The incident was mitigated when the change was fully rolled back, and all the connections were moved back to the previous functional gateway ring after validation.

How are we making incidents like this less likely or less impactful?

- MySQL engineering will improve operational practices by adding additional checks, testing and sign-offs procedures before switching the production gateway rings. (Estimated completion: May 2024)

- Azure Databricks engineering will introduce alerts to detect multiple databases being unavailable within a short span of time. (Estimated completion: May 2024)

How can customers make incidents like this less impactful?

- Azure Databricks customers should consider reviewing our best practices surrounding disaster recovery for Azure Databricks, see: https://learn.microsoft.com/azure/databricks/administration-guide/disaster-recovery
- As Azure Database for MySQL – Single Server and Azure Database for MariaDB are on the retirement path, customers are recommended to upgrade to Azure Database for MySQL – Flexible Server. The Flexible Server architecture has no shared gateways and is a single VM per tenant architecture. All planned maintenance in Flexible Server is scheduled in maintenance window defined by the end user for that server, which architecturally avoids such incidents. For details, see: https://learn.microsoft.com/azure/postgresql/flexible-server/overview
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/en-us/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/QS3K-BPZ

March 2024

Post Incident Review (PIR) - Network connectivity - Multiple services impacted in South Africa North and South Africa West

Tracking ID: VT60-RPZ

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/VT60-RPZ

What happened?

Between 10:33 UTC on 14 March 2024 and 11:00 UTC on 15 March 2024, customers using Azure services in the South Africa North and/or South Africa West regions may have experienced network connectivity failures, including extended periods of increased latency or packet drops when accessing resources. This incident was part of a broader continental issue, impacting telecom services to multiple countries in Africa.

The incident resulted from multiple concurrent fiber cable cuts that occurred on the west coast of Africa (specifically the WACS, MainOne, SAT3, and ACE cables) in addition to earlier ongoing cable cuts on the east coast of Africa (including the EIG, and Seacom cables). These cables are part of the submarine cable system that connect Africa's internet to the rest of the world, and service Microsoft's cloud network for our Azure regions in South Africa. In addition to the cable cuts, we later experienced impact reducing our backup capacity path, leading to congestion that impacted services.

Some customers may have experienced degraded performance including extended timeouts and/or service failures across multiple Microsoft services – while some customers may have been unaffected. Customer impact varied depending on the service(s), region(s), and configuration(s). Impacted downstream services included Azure API Management, Azure Application Insights, Azure Cognitive Services, Azure Communication Services, Azure Cosmos DB, Azure Databricks, Azure Event Grid, Azure Front Door, Azure Key Vault, Azure Monitor, Azure NetApp Files, Azure Policy, Azure Resource Manager, Azure Site Recovery, Azure SQL DB, Azure Virtual Desktop, Managed identities for Azure resources, Microsoft Entra Domain Services, Microsoft Entra Global Secure Access, Microsoft Entra ID, and Microsoft Graph. For service specific impact details, refer to the 'Health history' section of Azure Service Health within the Azure portal.

What went wrong, and why?

The Microsoft network is designed to support multiple failures to our Wide Area Network (WAN) capacity at any given point in time. Specifically, our regions in South Africa are connected via multiple diverse physical paths – both subsea and terrestrially within South Africa. The network is designed to support multiple failures and continue operating with only one single physical path. In this case, our South Africa Regions has four physically diverse subsea cable systems serving the country and the designed failure mode is that three out of four can fail with no impact to our customers.

Following news of geopolitical risks in the Red Sea, we ran internal simulations and capacity planning analysis. On 5 February, we initiated capacity additions to our African network. On 24 February, multiple cable cuts in the Red Sea impacted our east coast network capacity to Africa. This east coast capacity was unavailable, however there was no customer impact because of the built in redundancy.

Before our capacity additions from February had come online, on 14 March we experienced additional multiple concurrent fiber cable cuts, this time on the west coast of Africa – which further reduced the total network capacity for our Azure regions in South Africa. These cable cuts were due to a subsea seismic event (likely an earthquake and/or mudslide) which impacted multiple subsea systems – one of which is used by Microsoft. Additionally, after the west coast cable cuts had occurred, we experienced a line card optic failure on a Microsoft router inside the region that further reduced network headroom. Microsoft experiences hundreds of line card optic failures every day across the 500k+ devices that operate our network – such an event would normally have been invisible to our customers. However, the combination of concurrent cable cuts and this line card failure removed the necessary headroom on the failover path, which led to the congestion experienced.

This combination of events affected Azure services including Compute, Storage, Networking, Databases, and App Services – as well as Microsoft 365 services. While many customers leverage local instances of their services within the South Africa regions, there are some services that rely on API calls made to regions outside of South Africa. The reduced bandwidth to/from the South Africa regions, impacted these specific API calls and therefore impacted service availability and/or performance.

How did we respond?

The timeline that follows includes network availability figures, which represent the breadth of impact to our network capacity but may not represent the impact experienced by any specific customer or service.

- 3 February 2024 – News articles surfaced geopolitical risk to Red Sea subsea cable infrastructure.
- 5 February 2024 – Based on our internal simulations, we began the process of requesting capacity augments to Microsoft's west coast Africa network.

- 24 February 2024 – Multiple cable cuts in the Red Sea impacted east coast capacity (EIG, and Seacom cables), no impact to customers/services.
- 4 March 2024 – Local fiber providers began work on approved capacity augments.
- 14 March 2024 @ 10:02 UTC – Multiple cable cuts impacted west coast capacity (WACS + MAINONE + SAT3).
- 14 March 2024 @ 10:33 UTC – Customer impact began, as reduced capacity began to cause networking latency and packet drops, our on-call engineers began investigating. Network availability dropped as low as 77%.
- 14 March 2024 @ 11:55 UTC – Azure Front Door failed out of the region, to reduce inter-region traffic.
- 14 March 2024 @ 12:00 UTC – Individual cloud service teams began reconfigurations to optimize network traffic to reduce congestion.
- 14 March 2024 @ 15:44 UTC – After the combination of our mitigation efforts and the end of the business day in Africa, network traffic volume reduced – network availability rose above 97%.
- 14 March 2024 @ 16:25 UTC – We continued implementing traffic engineering measures to throttle traffic and reduce congestion – network availability rose above 99%.
- 15 March 2024 @ 06:00 UTC – As network traffic volumes increased, availability degraded, and customers began experiencing congestive packet loss – network availability dropped to 96%.
- 15 March 2024 @ 11:00 UTC – We shifted capacity from Microsoft's Edge in Lagos to increase headroom for South Africa, last packed drops observed on our WAN. While this effectively mitigated customer impact, we continued to monitor until additional capacity supported more headroom.
- 17 March 2024 @ 21:00 UTC – First tranche of emergency capacity came online.
- 18 March 2024 @ 02:00 UTC – Second tranche of emergency capacity came online, Azure Front Door brought back into our South Africa regions, incident declared mitigated.

How are we making incidents like this less likely or less impactful?

- We have added Wide Area Network (WAN) capacity to the region, in the form of a new physically diverse cable system with triple the capacity of pre-incident levels (Completed).
- We are reviewing our capacity augmentation processes to help accelerate urgent capacity additions when needed (Estimated completion: April 2024).
- We continue to work with our fiber providers to restore WAN paths after the cable cuts on the west coast of Africa (Estimated completion: April 2024) and on the east coast of Africa (Estimated completion: May 2024).
- We are evaluating adding a fifth WAN path between South Africa and the United Arab Emirates, to build even more resiliency to the rest of the world (Estimated completion: June 2024).
- We are increasingly shifting services to run locally from within our South Africa regions, to reduce dependencies on international regions where possible, including Exchange Online Protection (Estimated completion: June 2024).
- In the longer term, we are investing in WAN Gateways in Nigeria to improve our fault isolation and routing capabilities. (Estimated completion: December 2024)
- Finally, we are working to build out and activate Microsoft-owned fiber capacity to these regions, to reduce dependencies on local fiber providers. This includes investments in our own capacity on the new submarine cables going to Africa (specifically the Equiano, 2Africa East and West) which will exponentially increase capacity to serve our regions in South Africa. Importantly, this capacity will also be controlled by Microsoft – giving us more operational flexibility to add/change/move capacity in our WAN, versus relying on third-party telecom operators. These WAN fiber investments on new cable systems will land on the west coast of Africa (Estimated completion: December 2024) as well as on the east coast of Africa (Estimated completion: December 2025).

How can our customers and partners make incidents like this less impactful?

- During the incident, we advised customers who were able to fail out of these regions to consider doing so. Customers should consider a multi-region geodiversity strategy for mission-critical workloads, see https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application and https://learn.microsoft.com/azure/architecture/patterns/geodes
- Applications that use exponential-backoff in their retry strategy may have seen success, as an immediate retry during intervals of high packet loss may have also seen high packet loss, but a retry conducted during periods of lower loss would likely have succeeded. For more details on retry patterns, refer to https://learn.microsoft.com/azure/architecture/patterns/retry
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/en-us/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

*Y*ou can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/VT60-RPZ

January 2024

Post Incident Review (PIR) - Azure Resource Manager - Services impacted by ARM failures

Tracking ID: NKRF-1TG

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/NKRF-1TG

What happened?

Between 01:30 and 08:58 UTC on 21 January 2024, customers attempting to leverage Azure Resource Manager (ARM) may have experienced issues when performing resource management operations. This impacted ARM calls that were made via Azure CLI, Azure PowerShell and the Azure portal. While the impact was predominantly experienced in Central US, East US, South Central US, West Central US, and West Europe, impact may have been experienced to a lesser degree in other regions due to the global nature of ARM.

This incident also impacted downstream Azure services which depend upon ARM for their internal resource management operations – including Analysis Services, Azure Container Registry, API Management, App Service, Backup, Bastion, CDN, Center for SAP solutions, Chaos Studio, Data Factory, Database for MySQL flexible servers, Database for PostgreSQL, Databricks, Device Update for IoT Hub, Event Hubs, Front Door, Key Vault, Log Analytics, Migrate, Relay, Service Bus, SQL Database, Storage, Synapse Analytics, and Virtual Machines.

In several cases, data plane impact on downstream Azure services was the result of dependencies on ARM for retrieval of Role Based Access Control (RBAC) data (see: https://learn.microsoft.com/azure/role-based-access-control/overview). For example, services including Storage, Key Vault, Event Hub, and Service Bus rely on ARM to download RBAC authorization policies. During this incident, these services were unable to retrieve updated RBAC information and once the cached data expired these services failed, rejecting incoming requests in the absence of up-to-date access policies. In addition, several internal offerings depend on ARM to support on-demand capacity and configuration changes, leading to degradation and failure when ARM was unable to process their requests.

What went wrong and why?

In June 2020, ARM deployed a private preview integration with Entra Continuous Access Evaluation (see: https://learn.microsoft.com/entra/identity/conditional-access/concept-continuous-access-evaluation). This feature is to support continuous access evaluation for ARM, and was only enabled for a small set of tenants and private preview customers. Unbeknownst to us, this preview feature of the ARM CAE implementation contained a latent code defect that caused issues when authentication to Entra failed. The defect would cause ARM nodes to fail on startup whenever ARM could not authenticate to an Entra tenant enrolled in the preview.

On 21 January 2024, an internal maintenance process made a configuration change to an internal tenant which was enrolled in this preview. This triggered the latent code defect and caused ARM nodes, which are designed to restart periodically, to fail repeatedly upon startup. ARM nodes restart periodically by design, to account for automated recovery from transient changes in the underlying platform, and to protect against accidental resource exhaustion such as memory leaks.

Due to these ongoing node restarts and failed startups, ARM began experiencing a gradual loss in capacity to serve requests. Eventually this led to an overwhelming of the remaining ARM nodes, which created a negative feedback loop (increased load resulted in increased timeouts, leading to increased retries and a corresponding further increase in load) and led to a rapid drop in availability. Over time, this

impact was experienced in additional regions – predominantly affecting East US, South Central US, Central US, West Central US, and West Europe.

How did we respond?

At 01:59 UTC, our monitoring detected a decrease in availability, and we began an investigation. Automated communications to a subset of impacted customers began shortly thereafter and, as impact to additional regions became better understood, we decided to communicate publicly via the Azure Status page. By 04:25 UTC we had correlated the preview feature to the ongoing impact. We mitigated by making a configuration change to disable the feature. The mitigation began to rollout at 04:51 UTC, and ARM recovered in all regions except West Europe by 05:30 UTC.

The recovery in West Europe was slowed because of a retry storm from failed ARM calls, which increased traffic in West Europe by over 20x, causing CPU spikes on our ARM instances. Because most of this traffic originated from trusted internal systems, by default we allowed it to bypass throughput restrictions which would have normally throttled such traffic. We increased throttling of these requests in West Europe which eventually alleviated our CPUs and enabled ARM to recover in the region by 08:58 UTC, at which point the underlying ARM incident was fully mitigated.

The vast majority of downstream Azure services recovered shortly thereafter. Specific to Key Vault, we identified a latent bug which resulted in application crashes when latency to ARM from the Key Vault data plane was persistently high. This extended the impact for Vaults in East US and West Europe, beyond the vaults that opted into Azure RBAC.

- 20 January 2024 @ 21:00 UTC – An internal maintenance process made a configuration change to an internal tenant enrolled in the CAE private preview.
- 20 January 2024 @ 21:16 UTC – First ARM roles start experiencing startup failures, but no customer impact as ARM still has sufficient capacity to serve requests.
- 21 January 2024 @ 01:30 UTC – Initial customer impact due to continued capacity loss in several large ARM regions.
- 21 January 2024 @ 01:59 UTC – Monitoring detected additional failures in the ARM service, and on-call engineers began immediate investigation.
- 21 January 2024 @ 02:23 UTC – Automated communication sent to impacted customers started.
- 21 January 2024 @ 03:04 UTC – Additional ARM impact was detected in East US and West Europe.
- 21 January 2024 @ 03:24 UTC – Due to additional impact identified in other regions, we raised the severity of the incident, and engaged additional teams to assist in troubleshooting.
- 21 January 2024 @ 03:30 UTC – Additional ARM impact was detected in South Central US.
- 21 January 2024 @ 03:57 UTC – We posted broad communications via the Azure Status page.
- 21 January 2024 @ 04:25 UTC – The causes of impact were understood, and a mitigation strategy was developed.
- 21 January 2024 @ 04:51 UTC – We began the rollout of this configuration change to disable the preview feature.
- 21 January 2024 @ 05:30 UTC – ARM recovered in all regions except West Europe.
- 21 January 2024 @ 08:58 UTC – ARM recovered in West Europe, mitigating vast majority of customer impact beyond specific services who took more time to recover.
- 21 January 2024 @ 09:28 UTC – Key Vault recovered instances in West Europe by adding new scale sets to replace the VMs that had crashed due to the code bug.

How are we making incidents like this less likely or less impactful?

- Our ARM team have already disabled the preview feature through a configuration update. (Completed)
- We have offboarded all tenants from the CAE private preview, as a precaution. (Completed)
- Our Entra team improved the rollout of that type of per-tenant configuration change to wait for multiple input signals, including from canary regions. (Completed)
- Our Key Vault team has fixed the code that resulted in applications crashing when they were unable to refresh their RBAC caches. (Completed)
- We are gradually rolling out a change to proceed with node restart when a tenant-specific call fails. (Estimated completion: February 2024)
- Our ARM team will audit dependencies in role startup logic to de-risk scenarios like this one. (Estimated completion: February 2024)
- Our ARM team will leverage Azure Front Door to dynamically distribute traffic for protection against retry storm or similar events. (Estimated completion: February 2024)
- We are improving monitoring signals on role crashes for reduced time spent on identifying the cause(s), and for earlier detection of availability impact. (Estimated completion: February 2024)
- Our Key Vault, Service Bus and Event Hub teams will migrate to a more robust implementation of the Azure RBAC system that no longer relies on ARM and is regionally isolated with standardized implementation. (Estimated completion: February 2024)
- Our Container Registry team are building a solution to detect and auto-fix stale network connections, to recover more quickly from incidents like this one. (Estimated completion: February 2024)
- Finally, our Key Vault team are adding better fault injection tests and detection logic for RBAC downstream dependencies. (Estimated completion: March 2024).

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/NKRF-1TG

September 2023

Post Incident Review (PIR) – Services impacted by power, BIOS, and Virtual Machine issues – East US

Tracking ID: 2LZ0-3DG

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/2LZ0-3DG

What happened?

Between 07:24 and 19:00 UTC on 16 September 2023, a subset of customers using Virtual Machines (VMs) in the East US region experienced connectivity issues. This incident was triggered when a number of scale units within one of the datacenters in one of the Availability Zones lost power and, as a result, the nodes in these scale units rebooted. While the majority rebooted successfully, a subset of these nodes failed to come back online automatically. This issue caused downstream impact to services that were dependent on these VMs - including SQL Databases, Service Bus and Event Hubs. Impact varied by service and configuration:

- Virtual Machines were offline during this time. while recovery began at approximately 16:30 UTC, full mitigation was declared at 19:00 UTC.
- While the vast majority of zone-redundant Azure SQL Databases leveraging were not impacted, some customers using proxy mode connection may have experienced impact, due to one connectivity gateway not being configured with zone-resilience.
- SQL Databases with 'auto-failover groups' enabled were failed out of the region, incurring approximately eight hours of downtime prior to the failover completing.
- SQL Databases with 'active geo-replication' were able to self-initiate a failover to an alternative region manually to restore availability.
- The majority of SQL Databases were recovered no later than 19:00 UTC. Customers would have seen gradual recovery over time during mitigation efforts.
- Finally, non-zonal deployments of Service Bus and Event Hubs would have experienced a degradation. Zonal deployments of Service Bus and Event Hubs were unaffected.

What went wrong and why?

It is not uncommon for datacenters to experience an intermittent loss of power, and one of the ways we protect against this is by leveraging Uninterruptible Power Supplies (UPS). The role of the UPS is to provide stable power to infrastructure during short periods of power fluctuations, so that infrastructure does not fault or go offline. Although we have redundant UPS systems in place for added resilience, this incident was initially triggered by a UPS rectifier failure on a Primary UPS.

The UPS was connected to three Static Transfer Switches (STS) – which are designed to transfer power loads between independent and redundant power sources, without interruption. The STS is designed to remain on the primary source whenever possible, and to transfer back to it when stable power is available again. When the UPS rectifier failed, the STS successfully transferred to the redundant UPS – but then the primary UPS recovered temporarily, albeit in a degraded state. In this degraded state, the primary UPS is unable to provide stable power for the full load. So, after a 5-second retransfer delay, when the STS transferred from the redundant UPS back to the primary UPS, the primary UPS failed completely.

While the STS should then have transferred power back to the redundant UPS, the STS has logic designed to stagger these power transfers when there are multiple transmissions (to and from primary and redundant UPS) happening in a short period of time. This logic prevented the STS from transferring back to the redundant power, after the primary UPS failed completely, which ultimately caused a power loss to a subset of the scale units within the datacenter – at 07:24 UTC, for 1.9 seconds. This scenario of load transfers, to and from degraded UPS, over a short period of time, was not accounted for in the design. After 1.9 seconds, the load moved to the redundant source automatically for a final time. Our onsite datacenter team validated that stable power was feeding all racks immediately after the event, and verified that all devices were powered on.

Following the restoration of power, our SQL monitoring immediately observed customer impact, and automatic communications were sent to customers within 12 minutes. SQL telemetry also provided our first indication that some nodes were stuck during the boot up process. When compute nodes come online, they first check the network connectivity, then make multiple attempts to communicate with the preboot execution environment (PXE) server, to ensure that the correct network routing protocols can be applied. If the host cannot find a PXE server, it is designed to retry indefinitely until one becomes available so it can complete the boot process.

A previously discovered bug that applied to some of our BIOS software led to several hosts not retrying to connect to a PXE server, and remaining in a stuck state. Although this was a known issue, the initial symptoms led us to believe that there was a potential issue with the network and/or our PXE servers – troubleshooting these symptoms led to significant delays in correlating to the known BIOS issue. While multiple teams were engaged to help troubleshoot these issues, our attempts at force rebooting multiple nodes were not successful. As such, a significant amount of time was spent exploring additional mitigation options. Unbeknownst to our on call engineering team, these bulk reboot attempts were blocked by an internal approval process, which has been implemented as a safety measure to restrict the number of nodes that are allowed to be forced rebooted at one time. Once we understood all of the factors inhibiting mitigation, at around 16:30 UTC we proceeded to reboot the relevant nodes within the safety thresholds, which mitigated the BIOS issue successfully.

16

One of the mechanisms our platform deploys when VMs enter an unhealthy state is 'service healing' in which our platform automatically redeploys or migrates it to a healthy node. One of the prerequisites to initiate service healing requires a high percentage of nodes to be healthy – to ensure that, during a major incident, our self-healing systems do not exacerbate the situation. Once we had recovered past the safe threshold, the service healing mechanism initiated for the remainder of the nodes.

Throughout this incident, we did not have adequate alerting in place, and could not determine which specific VMs were impacted, because our assessment tooling relies on a heartbeat emitted from the compute nodes, which were stuck during the boot up process. Unfortunately, the time taken to understand the nature of this incident meant that communications were delayed. For customers using Service Bus and Event Hubs, this was multiple hours. For customers using Virtual Machines, this was multiple days. As such, we are investigating several communications related repairs, including why automated communications were not able to inform customers with impacted VMs in near real time, as expected.

How did we respond?

- 16 September 2023 @ 07:23 UTC - Loss of power to the three STSs.
- 16 September 2023 @ 07:24 UTC - All three downstream STSs fully re-energized.
- 16 September 2023 @ 07:33 UTC - Initial customer impact to SQL DB detected via monitoring.
- 16 September 2023 @ 07:34 UTC - Communications sent to Azure Service Health for SQL DB customers.
- 16 September 2023 @ 11:40 UTC - The relevant compute deployment team engaged to assist in rebooting nodes.
- 16 September 2023 @ 12:13 UTC - The infrastructure firmware team was engaged to troubleshoot the BIOS issues.
- 16 September 2023 @ 13:38 UTC - Multiple compute nodes attempted to be forcefully rebooted with no success.
- 16 September 2023 @ 15:30 UTC - SQL Databases with 'auto-failover groups' were successfully failed over.
- 16 September 2023 @ 15:37 UTC - Communications sent to Azure Service Health for Service Bus and Event Hub customers.
- 16 September 2023 @ 16:30 UTC - Safety thresholds blocking reboot attempts understood, successful batch rebooting begins.
- 16 September 2023 @ 16:37 UTC - Communications published to Azure Status page, in lieu of more accurate impact assessment.
- 16 September 2023 @ 19:00 UTC - All compute and SQL nodes successfully mitigated.
- 16 September 2023 @ 22:07 UTC - Once mitigation was validated, communications sent to Azure Service Health for SQL, Service Bus, and Event Hub customers.
- 19 September 2023 @ 04:10 UTC – Once VM impact was determined, communications sent to Azure Service Health for VM customers.

How are we making incidents like this less likely or less impactful?

- First and foremost, we have replaced the failed rectifier inside the UPS. (Completed)
- We are working with the manufacturer to perform a UPS rectifier failure analysis. (Estimated completion: October 2023)
- We are reviewing the status of STS automated transfer logic across all of our datacenters. (Estimated completion: October 2023)
- We are working to modify the STS logic to correct the transfer delay issue. (Estimated completion: December 2023)
- We have been deploying the fix for the BIOS issue as of January 2023 – we are expediting rollout. (Estimated completion: June 2024)
- We are improving our detection of stuck nodes for incidents of this class. (Estimated completion: October 2023)
- We are improving our automated mitigation of stuck nodes for incidents of this class. (Estimated completion: March 2024)
- We are improving the resiliency of our automated communication system for incidents of this class. (Estimated completion: October 2023)
- We are reviewing the status of STS automated transfer in all our sites. (Estimated completion: October 2023)
- For the issue surrounding Multi-AZ Azure SQL Databases using a proxy mode connection, the fix was already underway before this incident and has since been deployed. (Completed)

How can customers make incidents like this less impactful?

- Consider using Availability Zones (AZs) to run your services across physically separate locations within an Azure region. To help services be more resilient to datacenter-level failures like this one, each AZ provides independent power, networking, and cooling. Many Azure services support zonal, zone-redundant, and/or always-available configurations: https://docs.microsoft.com/azure/availability-zones/az-overview
- For mission-critical workloads, customers should consider a multi-region geodiversity strategy to avoid impact from incidents like this one that impacted a single region: https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application/ and https://learn.microsoft.com/azure/architecture/patterns/geodes
- We encourage customers to review and follow our guidance and best practices around Azure SQL Database disaster recovery – practice disaster drills to ensure that your application can handle the cross-region failover gracefully: https://learn.microsoft.com/azure/azure-sql/database/disaster-recovery-guidance

- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/azure/architecture/framework/resiliency
- Consider ensuring that the right people in your organization will be notified about any future service issues by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/2LZ0-3DG

August 2023

Post Incident Review (PIR) - Services impacted after power/cooling issue - Australia East

Tracking ID: VVTQ-J98

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/VVTQ-J98

What happened?

Starting at approximately 10:30 UTC on 30 August 2023, customers may have experienced issues accessing or using Azure, Microsoft 365 and Power Platform services. This incident was triggered by a utility power sag at 08:41 UTC on 30 August 2023, which impacted one of the three Availability Zones of the Australia East region. This power sag tripped a subset of the cooling system chiller units offline and, while working to restore cooling, temperatures in the datacenter increased to levels above operational thresholds. We powered down a small subset of selected compute and storage scale units, both to lower temperatures and to prevent damage to hardware. Although the vast majority of services recovered by 22:40 UTC on 30 August 2023, full mitigation was not until 20:00 UTC on 3 September 2023 – as some services experienced prolonged impact, predominantly as a result of dependencies on recovering subsets of Storage, SQL Database, and/or Cosmos DB services.

Multiple Azure services were impacted by this incident – including Azure Active Directory (AAD), Azure Active Directory B2C, Azure Active Directory Conditional Access, Azure Active Directory Connect Health, Azure Active Directory MyApps, Azure Activity Logs & Alerts, Azure API Management, Azure App Service, Azure Application Insights, Azure Arc enabled Kubernetes, Azure API for FHIR, Azure Backup, Azure Batch, Azure Chaos Studio, Azure Container Apps, Azure Container Registry, Azure Cosmos DB, Azure Databricks, Azure Data Explorer, Azure Data Factory, Azure Database for MySQL flexible servers, Azure Database for PostgreSQL flexible servers, Azure Digital Twins, Azure Device Update for IoT Hub, Azure Event Hubs, Azure ExpressRoute, Azure Health Data Services, Azure HDInsight, Azure IoT Central, Azure IoT Hub, Azure Kubernetes Service (AKS), Azure Logic Apps, Azure Log Analytics, Azure Log Search Alerts, Azure NetApp Files, Azure Notification Hubs, Azure Redis Cache, Azure Relay, Azure Resource Manager (ARM), Azure Role Based Access Control (RBAC), Azure Search, Azure Service Bus, Azure Service Fabric, Azure SQL Database, Azure Storage, Azure Stream Analytics, Azure Virtual Machines, Microsoft Purview, and Microsoft Sentinel.

What went wrong and why?

Starting at approximately 08:41 UTC on 30 August 2023, a utility voltage sag was caused by a lightning strike on electrical infrastructure approximately 18 miles from the impacted Availability Zone of the Australia East region. The voltage sag caused cooling system chillers for multiple datacenters to shut down. While some chillers automatically restarted, 13 failed to restart and required manual intervention. To do so, the onsite team accessed the datacenter rooftop facilities, where the chillers are located, and proceeded to sequentially restart chillers moving from one datacenter to the next. By the time the team reached the final five chillers requiring a manual restart, the water inside the pump system for these chillers (chilled water loop) had reached temperatures that were too high to allow them to be restarted. In this scenario, the restart is inhibited by a self-protection mechanism that acts to prevent damage to the chiller that would occur by processing water at the elevated temperatures. The five chillers that could not be restarted supported cooling for the two adjacent data halls which were impacted in this incident.

The two impacted data halls require at least four chillers to be operational. Before the voltage sag, our cooling capacity for these halls consisted of seven chillers, with five chillers in operation and two chillers in standby. At 10:30 UTC some networking, compute, and storage infrastructure began to shutdown automatically as data hall temperatures increased, impacting service availability. At 11:34 UTC, as temperatures continued to increase, our onsite datacenter team began a remote shutdown of remaining networking, compute, and storage infrastructure in the impacted data halls to protect data durability, infrastructure health, and address the thermal runaway. This shutdown allowed the chilled water loop to return to a safe temperature which allowed us to restart the chillers. This shutdown of infrastructure resulted in a further reduction of service availability for this Availability Zone.

The chillers were successfully brought back online at 12:12 UTC, and data hall temperatures returned to operational thresholds by 13:30 UTC. Power was then restored to the affected infrastructure and a phased process to bring the infrastructure back online commenced. All power to infrastructure was restored by 15:10 UTC. Once all networking and storage infrastructure had power restored, dependent compute scale units were then also returned to operation. As the underlying compute and storage scale units came online, dependent Azure services started to recover, but some services experienced issues coming back online.

From a storage perspective, seven storage scale units were impacted – five standard storage scale units, and two premium storage scale units. Availability impact to affected storage accounts began at 10:30 UTC as hardware shut down in response to elevated data hall temperatures. This was most impactful to storage accounts configured with the default local redundant storage (LRS), which is not resilient to a zonal failure. Accounts configured as zonally redundant (ZRS) remained 100% available, and accounts configured as geographically redundant (GRS) were eligible for customer-managed account failover. After power restoration, storage nodes started coming back online from 15:25 UTC. Four scale units required engineer intervention to check and reset some fault detection logic – the combination of investigation and management tooling performance problems delayed restoration of availability for these scale units. We also identified that some automation was incorrectly marking some already recovered nodes as unhealthy, which slowed storage recovery efforts. By 20:00 UTC, 99% of storage accounts had recovered. Restoring availability for the remaining <1% of storage accounts took more time due to hardware troubleshooting and replacement required on a small number of storage nodes in a single scale unit. Even identifying problematic hardware in the storage nodes took an extended period of time, as the nodes were offline and therefore not able to provide diagnostics. By 01:00 UTC on 31 August, availability was restored for all except a handful of storage accounts, with complete availability restored by 07:00 UTC on 1 September.

From a SQL Database perspective, database capacity in a region is divided into tenant rings. The Australia East region includes hundreds of rings, and each ring consists of a group of VMs (10-200) hosting a set of databases. Rings are managed by Azure Service Fabric to provide availability in cases of VM, network or storage failures. When infrastructure was powered down, customers using zone-redundant Azure SQL Database did not experience any downtime, except for a small subset of customers using proxy mode connection, due to one connectivity gateway not being configured with zonal-resilience. The fix for this issue was already being rolled out, but had not yet deployed to the Australia East region. As infrastructure was powered back on, all tenant rings except one came back online and databases became available to customers as expected. However, one ring remained impacted even after Azure Compute became available. In this ring, 20 nodes did not come back online as expected, so databases on these nodes continued to experience unavailability. As a result of Service Fabric attempting to move databases to healthy nodes, other databases on this ring experienced intermittent availability issues as a side-effect of the overall replica density and unhealthy nodes. The recovery involved first moving all the databases from unhealthy nodes to healthy nodes. All remote storage (general purpose) databases were successfully recovered by this move, but databases using local storage (business critical) only recovered as their underlying nodes recovered. All databases on unhealthy nodes were recovered by 11:00 UTC on 31 August. Since the health and capacity of this final ring did not completely recover, we decided to move all databases out of the ring, which extended the overall recovery time but did not negatively impact customer availability. During this extended recovery, most customers were not experiencing any issues but it was important to move all databases out of this unhealthy ring to prevent any potential impact. The operation of moving all databases out of this ring was completed at 20:00 UTC on 3 September. During this incident, customers who had 'active geo-replication' setup were able to failover manually to restore availability. For customers who have 'auto-failover groups' enabled, we did not execute automatic failover – our automatic failover policy was not initiated for the region, due to an incorrect initial assessment of the impact severity to SQL Database.

From a Cosmos DB perspective, zone resilient accounts and those with multi-region writes remained operational during the incident, transparently serving requests from a different zone or region, respectively. However, accounts not configured for AZ or multi-region writes experienced full or partial loss of availability, due to the infrastructure that was powered down. Multi-region accounts with single region write eligible for failover (those with Service Managed Failover enabled) were failed over to their alternate regions to restore availability. These were initiated at 12:07 UTC, 33 minutes after decision to power down scale units. The reason for this delay was to identify and failover the Cosmos DB control plane system resources – in retrospect this delay was unnecessary, as the Cosmos DB control plane was already fully zone-resilient. 95% of database accounts were failed over within 35 minutes by 12:42 UTC, and all eligible accounts were failed over by 16:13 UTC on 30 August. Accounts that were not eligible for failover had service restored to partitions only once the dependent storage and compute were restored.

From an Azure Kubernetes Service (AKS) perspective, the service experienced a loss of compute for the AKS control plane for Australia East as well as data access loss to SQL Database. The AKS control plane underlay is deployed across multiple availability zones. AKS uses Azure SQL Database for its operation queue which is used for Create/Read/Update/Delete (CRUD) activities. Although scheduled to be converted, the SQL Database in Australia East was not configured with AZ resiliency selection, leaving it unavailable during the incident period. In addition, AKS services in the Australia Southeast region depended on this same database, causing an AKS incident for CRUD activities in that region also. Existing customer workloads running on AKS clusters in either region should not have been impacted by the downtime, as long as they did not need to access the AKS resource provider for scaling or other CRUD activities. As the SQL Database recovered, service was restored without any other mitigation required.

From an Azure Resource Manager (ARM) perspective, the impact on customers was the result of degradation in Cosmos DB. This degradation impacted ARM between 10:45 UTC and 12:25 UTC and resulted in ARM platform availability for the Australia East region dropping from ~99.999% to (at its lowest) 88%, with a 62% success rate for write operations. For data consistency reasons, write operations are required to be sent to the associated Cosmos DB regional replica for a given resource group. While the migration to our next generation zonally-redundant storage architecture is still ongoing, it has not been completed and as a result this region is not yet

leveraging fully zonally redundant storage for ARM. This meant that for the duration of the incident, customers worldwide trying to manage resources whose resource groups were homed in Australia East saw increased error rates (and this manifested in a small impact to global platform availability until 15:00 UTC).

How did we respond?

- 30 August 2023 @ 08:41 UTC – Voltage sag occurred on utility power line
- 30 August 2023 @ 08:43 UTC – 13 chillers failed to restart automatically
- 30 August 2023 @ 08:51 UTC – Remote resets on chillers commenced
- 30 August 2023 @ 09:09 UTC – Team arrived at first group of chillers for manual restarts
- 30 August 2023 @ 09:18 UTC – Team arrived at second group of chillers for manual restarts
- 30 August 2023 @ 09:42 UTC – Team arrived at third group of chillers for manual restarts
- 30 August 2023 @ 09:45 UTC – Team arrived at the final group of chillers which could not be restarted
- 30 August 2023 @ 10:30 UTC – Initial impact from automated infrastructure shutdown
- 30 August 2023 @ 10:47 UTC – Cosmos DB Initial impact detected via monitoring
- 30 August 2023 @ 10:48 UTC – First automated communications sent to Azure Service Health
- 30 August 2023 @ 11:30 UTC – Initial communications posted to public Azure Status page
- 30 August 2023 @ 11:34 UTC – Decision made to shutdown impacted infrastructure
- 30 August 2023 @ 11:36 UTC – All subscriptions in Australia East sent portal communications
- 30 August 2023 @ 12:07 UTC – Failover initiated for eligible Cosmos DB accounts
- 30 August 2023 @ 12:12 UTC – Five chillers manually restarted
- 30 August 2023 @ 13:30 UTC – Data hall temperature normalized
- 30 August 2023 @ 14:10 UTC – Safety walkthrough completed for both data halls
- 30 August 2023 @ 14:25 UTC – Decision made to start powering up hardware in the two affected data halls
- 30 August 2023 @ 15:10 UTC – Power restored to all hardware
- 30 August 2023 @ 15:25 UTC – Storage, networking, and compute infrastructure started coming back online after power restoration
- 30 August 2023 @ 15:30 UTC – Identified three specific storage scale units still experiencing fault codes
- 30 August 2023 @ 16:00 UTC – 35% of VMs recovered / Began manual recovery efforts for three remaining storage scale units
- 30 August 2023 @ 16:13 UTC – Account failover completed for all Cosmos DB accounts
- 30 August 2023 @ 17:00 UTC – All but one SQL Database tenant ring had recovered
- 30 August 2023 @ 19:20 UTC – 90% of VMs recovered
- 30 August 2023 @ 19:29 UTC – Successfully recovered all premium storage scale units
- 30 August 2023 @ 20:00 UTC – 99% of storage accounts were back online
- 30 August 2023 @ 22:35 UTC – Standard storage scale units were recovered, except for one scale unit
- 30 August 2023 @ 22:40 UTC – 99% of VMs recovered
- 31 August 2023 @ 04:04 UTC – Restoration of Cosmos DB accounts to Australia East initiated
- 31 August 2023 @ 04:43 UTC – Final Cosmos DB cluster recovered, restoring all traffic for accounts that were not failed over to alternate regions
- 31 August 2023 @ 05:00 UTC – 100% of VMs recovered
- 1 September 2023 @ 06:40 UTC – Successfully recovered all standard storage scale units
- 3 September 2023 @ 20:00 UTC – Final SQL Database tenant ring evacuated and all customer databases online

How are we making incidents like this less likely or less impactful?

- Incremental load increases over time in the Availability Zone resulted in a chiller operating configuration that was susceptible to a timing defect in the Chiller Management System. We have de-risked this failure to restart due to voltage fluctuations, by implementing a change to the control timing logic on the Chiller Management System (Completed).
- Emergency Operation Procedure for manual restarts of simultaneous chiller failures has changed from an ‘adjacent’ sequence to a data hall ‘load based’ sequence to ensure all impacted data halls have partial cooling, to slow thermal runaway while full cooling is being restored (Completed).
- Following this incident, as a temporary mitigation, we increased technician staffing levels at the datacenter to be prepared to execute manual restart procedures of our chillers prior to the change to the Chiller Management System to prevent restart failures. Based on our incident analysis the staffing levels at the time would have been sufficient to prevent impact if a ‘load based' chiller restart sequence had been followed, which we have since implemented (Completed).
- Datacenter staffing levels published in the Preliminary PIR only accounted for “critical environment” staff onsite. This did not characterize our total datacenter staffing levels accurately. To alleviate this misconception, we made a change to the preliminary public PIR posted on the Status History page.
- Our Storage team has identified several optimizations in our large scale recovery process which will help to reduce time to mitigate. This includes augmenting data provided in our incidents to enable quicker decision making, and updates to our troubleshooting guides (TSGs) that enable faster execution (Estimated completion: December 2023).
- Our Azure Service Fabric team is working to improve reliability of SQL Database tenant ring recovery. (Estimated completion: December 2023).
- Our SQL Database team is reviewing our ‘auto-failover group’ trigger criteria, to ensure that failovers can happen within the expected timeframe. (Estimated completion: October 2023).

- Our SQL Database team is upgrading internal tooling to enable mass migration of databases. (Estimated completion: December 2023).
- Our Cosmos DB team is working to optimize Service Managed Failover for single region write accounts to reduce time to mitigate (Estimated completion: November 2023).
- Our AKS team is immediately converting all operation queue SQL Database databases to be zone redundant (Estimated completion: September 2023).
- Our AKS team is also replacing all cross-region SQL Database queue usage with Service Bus queues that are zone redundant (Estimated completion: September 2023).
- Our ARM team will complete its storage layer migration to the next generation, zonally redundant architecture (Estimated completion: December 2023).
- Our incident management team is exploring ways to harden our readiness, process, and playbook surrounding power down scenarios (Estimated completion: October 2023)

How can customers make incidents like this less impactful?

- Consider using Availability Zones (AZs) to run your services across physically separate locations within an Azure region. To help services be more resilient to datacenter-level failures like this one, each AZ provides independent power, networking, and cooling. Many Azure services support zonal, zone-redundant, and/or always-available configurations: https://docs.microsoft.com/azure/availability-zones/az-overview
- For mission-critical workloads, customers should consider a multi-region geodiversity strategy to avoid impact from incidents like this one that impacted a single region: https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application/ and https://learn.microsoft.com/azure/architecture/patterns/geodes
- Consider which are the right storage redundancy options for your critical applications. Zone redundant storage (ZRS/GZRS) remains available throughout a zone localized failure, like in this incident. Geo-redundant storage (GRS) enables account level failover in case the primary region endpoint becomes unavailable: https://learn.microsoft.com/azure/storage/common/storage-redundancy
- Consider the relevant guidance for recovering your SQL Database databases during disaster recovery scenarios: https://learn.microsoft.com/azure/azure-sql/database/disaster-recovery-guidance
- Consider the relevant guidance for achieving high availability with Azure Cosmos DB https://learn.microsoft.com/azure/cosmos-db/high-availability
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues – by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/VVTQ-J98

July 2023

Post Incident Review (PIR) - Azure Monitor - Logs data access issues

Tracking ID: XMGF-5Z0

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/XMGF-5Z0

What happened?

Between 23:15 UTC on 6 July 2023 and 09:00 UTC on 7 July 2023, a subset of data for Azure Monitor Log Analytics and Microsoft Sentinel failed to ingest. Additionally, platform logs gathered via Diagnostic Settings failed to route some data to customer destinations such as Log Analytics, Storage, Event Hub and Marketplace. These failures were caused by a deployment of a service within Microsoft, with a bug that caused a much higher than expected call volume that overwhelmed the telemetry management control plane. Customers in all regions experienced impact.

Security Operations Center (SOC) functionality in Sentinel may have been impacted. Queries against impacted tables with date range listed above, inclusive of the logs data that we failed to ingest, might have returned partial or empty results. This includes analytics (detections), hunting queries, workbooks with custom queries, and notebooks. In cases where Event or Security Event tables were impacted, incident investigations of a correlated incident may have showed partial or empty results.

What went wrong and why?

A code deployment for the Azure Container Apps service was started on 3 July 2023 via the normal Safe Deployment Practices (SDP), first rolling out to Azure canary and staging regions. This version contained a misconfiguration that blocked the service from starting normally. Due to the misconfiguration, the service bootstrap code threw an exception, and was automatically restarted. This caused the bootstrap service to be stuck in a loop where it was being restarted every 5 to 10 seconds. Each time the bootstrap service was restarted, it provided configuration information to the telemetry agents also installed on the service hosts. Each time the configuration information was sent to the telemetry hosts, they interpreted this as a configuration change, and therefore they also automatically exited their current process and restarted as well. Three separate instances of the agent telemetry host, per application host, were now also restarting every 5 to 10 seconds.

Upon each startup of the telemetry agent, the agent immediately contacted the telemetry control plane to download the latest version of the telemetry configuration. Normally this is an action that would take place one time every several days, as this configuration would be cached on the agent. However, as the deployment of the Container Apps service progressed, several hundred hosts now had their telemetry agents requesting startup configuration information from the telemetry control plane every 5-10 seconds. The Container Apps team detected the fault in their deployment on 6 July 2023, stopped the original deployment before it was released to any production regions, and started a new deployment of their service in the canary and staging regions to correct the misconfiguration.

However, the aggregate rate of requests from the services that received the build with the misconfiguration exhausted capacity on the telemetry control plane. The telemetry control plane is a global service, used by services running in all public regions of Azure. As capacity on the control plane was saturated, other services involved in ingestion of telemetry, such as the ingestion front doors and the pipeline services that route data between services internally, began to fail as their operations against the telemetry control plane were either rejected or timed out. The design of the telemetry control plane as a single point of failure is a known risk, and investment to eliminate this risk has been underway in Azure Monitor to design this risk out of the system.

How did we respond?

The impact on the telemetry control plane grew slowly and did not create problems that were detected until 12:30 UTC on 6 July 2023. When the issues were detected, the source of the additional load against the telemetry control plane was not known, but the team suspected additional load had been created against the control plane and took these actions:

- 6 July 2023 @ 14:53 UTC – Internal incident bridge created.
- 6 July 2023 @ 15:56 UTC – ~500 instances of garbage collector service were removed, to reduce load on telemetry control plane
- 6 July 2023 @ 16:09 UTC – First batch of Node Diagnostics servers were removed, to reduce load on telemetry control plane. This process of removing this type of server continued over the next 10 hours.
- 6 July 2023 @ 20:20 UTC – Source of anomalously high traffic was identified, and the responsible team was paged to assist.
- 6 July 2023 @ 23:00 UTC – IP address blocks deployed, to prevent anomalous traffic from hitting telemetry control plane.
- 6 July 2023 @ 23:15 UTC – External customer impact started, as cached data started to expire.
- 7 July 2023 @ 01:30 UTC – Three additional clusters were added to telemetry control plane to handle additional load, and we began restarting existing clusters to clear backlogged connections.
- 7 July 2023 @ 02:19 UTC – Initial customer notification was posted to the Azure Status page, to acknowledge the incident was being investigated while we worked to identify which specific subscriptions were impacted.
- 7 July 2023 @ 02:45 UTC – An additional three clusters were added to telemetry control plane.
- 7 July 2023 @ 06:15 UTC – Targeted customer notifications sent via Azure Service Health to customers with impacted subscriptions (sent on Tracking ID XMGF-5Z0).
- 7 July 2023 @ 09:00 UTC – Incident declared mitigated, as call error rate and call latency against control plane APIs stabilized at typical levels.

How are we making incidents like this less likely or less impactful?

We know customer trust is earned and must be maintained, not just by saying the right thing but by doing the right thing. Data retention is a fundamental responsibility of the Microsoft cloud, including every engineer working on every cloud service. We have learned from this incident and are committed to the following improvements:

- We have ensured that our telemetry control plane services are now running with additional capacity (Completed)
- We have created additional alerting on certain metrics that indicate critical, unusual failure patterns in API calls (Completed)
- We will be adding new positive caching and negative caching to the control plane, to reduce load on backing store (Estimated completion: September 2023)
- We are putting in place additional throttling and circuit breaker patterns to our core telemetry control plane APIs (Estimated completion: September 2023)
- In the longer term, we are creating isolation between internal-facing and external-facing services using the telemetry control plane (Estimated completion: December 2023)

How can customers make incidents like this less impactful?

Note that the Azure Monitoring Agent (AMA) provides more advanced collection and ingestion resilience capabilities (such as caching, buffering and retries) than the legacy Microsoft Monitoring Agent (MMA). Customers who have not yet completed their migration from MMA to AMA, would benefit from accelerating and completing the migration - for use cases required by them and supported in AMA. For more details: https://learn.microsoft.com/azure/azure-monitor/agents/azure-monitor-agent-migration

Customers using Microsoft Sentinel can consider the following compensating steps:

- Identify high priority assets, log sources that cover those assets, and detection or hunting logic normally applied to those assets and logs.
- If possible, run one-time queries at the source, for the indicated date range where ingestion was impacted - based on the prioritized assets, logs, and logic. It may also be useful to run the same queries for week prior to the incident, compare, and look for differences.
- Alternatively, to the extent the collection architecture allows for it, re-ingest data from the source into Sentinel, and run those one-time queries in Sentinel.

Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/XMGF-5Z0

Post Incident Review (PIR) - Networking fiber cut - West Europe

Tracking ID: FVHB-188

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/FVHB-188

What happened?

Between approximately 07:22 UTC and 16:00 UTC on 5 July 2023, Azure customers using the West Europe region may have experienced packet drops, timeouts, and/or increased latency. This impact resulted from a fiber cut caused by severe weather conditions in the Netherlands. The West Europe region has multiple datacenters and is designed with four independent fiber paths for the traffic that flows between datacenters. In this incident, one of the four major paths was cut, which resulted in congestive packet loss when traffic on the remaining links exceeded their capacity.

Downstream Azure services dependent on this intra-region network connectivity were also impacted – including Azure App Services, Azure Application Insights, Azure Data Explorer, Azure Database for MySQL, Azure Databricks (which also experienced impact in North Europe, as a result of a control plane dependency), Azure Digital Twins, Azure HDInsight, Azure Kubernetes Service, Azure Log Analytics, Azure Monitor, Azure NetApp Files, Azure Resource Graph, Azure Site Recovery, Azure Service Bus, Azure SQL DB, Azure Storage, and Azure Virtual Machines – as well as subsets of Microsoft 365, Microsoft Power Platform, and Microsoft Sentinel services.

What went wrong and why?

Due to a fiber cut caused by severe weather conditions in the Netherlands, 25% of network links between two campuses of West Europe datacenters became unavailable. These links were already running at higher utilization than our design target, and there was a capacity augment project in progress to address this. Due to a previous incident related to this capacity augment on 16 June 2023 (Tracking ID VLB8-1Z0), the augment work was proceeding with extreme caution and was still in progress when the fiber cut occurred.

As a result of the fiber cut and the higher utilization, congestion increased to a point where intermittent packet drops occurred in many of the intra-region paths. This primarily impacted network traffic between Availability Zones within the West Europe region, not traffic to and from the region itself. As a result of this interruption, Azure services that rely on internal communications with other services within the region may have experienced degraded performance, manifesting in the issues described above.

How did we respond?

Network alerting services indicated a fiber cut at 07:22 UTC and a congestion alert triggered at 07:46 UTC. Our networking on-call engineers engaged and began to investigate. Two parallel workstreams were spun up to mitigate impact:

The first workstream focused on reducing traffic in the region and balancing it across the remaining links. This balancing activity requires a detailed before-and-after traffic simulation to ensure safety, and these simulations were initiated as a first step. At 10:00 UTC we initiated throttling and migration of internal service traffic away from the region. We also started work on rebalancing traffic away from congested links. As a result of these activities, by 14:52 UTC packet drops were reduced significantly, by 15:30 UTC many internal and external services saw signs of recovery, and by 16:00 UTC packet drops had returned to pre-incident levels. We continued to work on reducing high link utilization and by 16:21 UTC the rebalancing activity was completed.

The second workstream focused on repairing the impacted links, in partnership with our dark fiber provider in the Netherlands. These cable repairs took longer than expected since access to the impacted area was hindered by the weather and hazardous working conditions. Partial restoration was confirmed by 19:30 UTC, and full restoration was confirmed by 20:50 UTC. While this restored the

network capacity between datacenters, we continued to monitor our network infrastructure and capacity before declaring the incident mitigated at 22:45 UTC.

How are we making incidents like this less likely or less impactful?

- We have repaired the impacted networking links, in partnership with our dark fiber provider in the Netherlands. (Completed)
- Within 24 hours of the incident being mitigated we brought additional capacity online, on the impacted network path. (Completed)
- Within a week of the incident, we are 90% complete with our capacity augments that will double capacity in our West Europe region to bring utilization within our design targets. (Estimated completion: July 2023)
- As committed in a previous Post Incident Review (PIR), we are working towards auto-declaring regional incidents to ensure customers get notified more quickly (Estimated completion: August 2023).

How can customers make incidents like this less impactful?

- During the incident, we advised customers who were able to fail out of the West Europe region to consider doing so. Customers should consider a multi-region geodiversity strategy for mission-critical workloads, to avoid impact from incidents like this one that impacted a single region: https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application/ and https://learn.microsoft.com/azure/architecture/patterns/geodes
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/FVHB-188

June 2023

Post Incident Review (PIR) - Networking connectivity issues - West Europe

Tracking ID: VLB8-1Z0

Watch our 'Azure Incident Retrospective' video about this incident: https://aka.ms/AIR/VLB8-1Z0

What happened?

Between 02:34 UTC and 07:25 UTC on 16 June 2023, a network issue caused excessive packet loss that affected traffic entering or leaving the West Europe region and, to a lesser extent, traffic between Availability Zones inside the West Europe region. Resources hosted in this region may have experienced availability failures, low throughput, or increased latencies. The traffic loss rate peaked at 10% for short periods of time during the incident.

Downstream Azure services dependent on this network connectivity were also impacted – including Azure API Management, Azure App Services, Azure Application Insights, Azure Automation, Azure Container Apps, Azure Container Registry, Azure Cosmos DB, Azure Data Explorer, Azure Data Factory, Azure Database for MySQL, Azure Database for PostgreSQL, Azure Databricks (which also experienced impact in North Europe, as a result of a control plane dependency), Azure DevOps, Azure Digital Twins, Azure Event Grids, Azure HDInsight, Azure IoT Hub, Azure Key Vault, Azure Kubernetes Service, Azure Logic Apps, Azure Monitor, Azure NetApp Files, Azure Resource Graph, Azure Resource Manager, Azure Service Bus, Azure Site Recovery, Azure SQL Database, Azure Spatial Anchors, Azure Synapse, Azure Update Management Center, Azure Virtual Desktop, Azure Virtual Machines, and Azure Web PubSub – as well as subsets of Microsoft 365, Microsoft Intune, Microsoft Purview, Microsoft Sentinel, and Microsoft Power Platform services.

What went wrong and why?

The Azure network is comprised of millions of links between routers, with many redundant links to cope with failures. As a faulty link can cause packet loss or corruption, Azure has network automation systems that continually monitor these links for health. When our automation systems detect a link is unhealthy, the system confirms it is safe to remove the link from service; issues commands to the routers to shut down the link; and then issues a request to datacenter staff to repair or replace the faulty link.

A network architecture detail relevant to this incident is that the routers carrying traffic between Availability Zones and the Wide Area Network in West Europe are connected by groups of multiple physical links bundled together into what are called Link Aggregation Groups (LAGs or port-channels). To keep traffic balanced across the physical links, if one or more links in a LAG fail it is desirable to turn

off the entire LAG.

Shortly before the start time of this incident, a new network topology description was added into the network automation systems that manage the West Europe region. This new topology included links that were about to be added to the network to increase the capacity in West Europe, but the physical links had not yet been connected or turned on. This network topology description incorrectly listed these links as being 'Production'. The network automation systems detected LAGs that contained working physical links as well as the non-working 'Production' links. Then, as designed, network automation began issuing commands consistent with standard mitigations to turn off the LAGs, to prevent impact due to potential link imbalance.

Although regions are built with extensive redundancy, enough links were taken out of service to cause congestion and impact to customer traffic. As soon as the network automation systems reached our operational safety thresholds, they determined it was unsafe to continue taking links out of service and stopped their activity, as designed. This safety mechanism prevented the impact from worsening.

16 **How did we respond?**

Packet loss detection alerts fired at 03:03 UTC and notified our on-call engineers that help was needed to recover the network. Engineers scanned the network to identify links that were incorrectly removed from service and distinguish them from links that were correctly removed from service for being unhealthy. Although these mitigation efforts were in progress by 04:45 UTC, the packet loss worsened from 02:34 to 05:45 as traffic in the region increased with the workday starting in the EMEA region.

Engineers placed the healthy links back into service and ensured that the network automation would not remove them again. From 05:45 UTC, the network recovered progressively throughout the remaining incident period, as the links were restored. The network was substantially recovered by 06:12 UTC, and loss rates returned to normal by 07:25 UTC. Other services dependent on the network recovered shortly thereafter. This capacity augment project in West Europe is halted and will only be resumed after a thorough review and update of the current process.

How are we making incidents like this less likely or less impactful?

The two key factors that contributed to this incident were (1) a capacity augment project carried out where the state of the new to-be-added links was set to "Production" within our network automation system, and our system detected these links as inoperative and initiated mitigation actions; and (2) the system that acts as a safety gate for the auto-mitigation actions estimated the degree of impact incorrectly, and allowed the auto-mitigation systems to take down too much capacity on the network, which led to congestion and packet drops. As a result of these learnings, our repair items include the following:

- We have enhanced our alerts to accelerate packet loss detection and decrease time to detect (Completed)
- We are enhancing our roll-back capability for our on-call engineers to speed up recovery (Estimated completion: August 2023).
- We will auto-declare regional incidents to ensure customers get notified more quickly (Estimated completion: August 2023).
- We are making improvements to the procedure that was used for this specific capacity augment scenario to ensure non-working links are not set to an In-Production state (Estimated completion: September 2023).
- We are making the safety gates used by our auto-mitigations more stringent with respect to network utilization and capacity headroom (Estimated completion: September 2023).
- We are making improvements to the procedure that was used for this specific capacity augment scenario to ensure non-working links are not set to a Production state (Estimated completion: October 2023).

How can customers make incidents like this less impactful?

- Consider a multi-region geodiversity strategy for your mission-critical workloads, to avoid impact from incidents like this one that impacted a single region: https://learn.microsoft.com/training/modules/design-a-geographically-distributed-application/ and https://learn.microsoft.com/azure/architecture/patterns/geodes
- Applications that use exponential-backoff in their retry strategy may have seen success, as an immediate retry during the short intervals of high packet loss may have also seen high packet loss, but a retry conducted during periods of lower loss would have likely succeeded. For more details on retry patterns, refer to https://learn.microsoft.com/azure/architecture/patterns/retry
- More generally, consider evaluating the reliability of your applications using guidance from the Azure Well-Architected Framework and its interactive Well-Architected Review: https://docs.microsoft.com/azure/architecture/framework/resiliency
- Finally, consider ensuring that the right people in your organization will be notified about any future service issues - by configuring Azure Service Health alerts. These can trigger emails, SMS, push notifications, webhooks, and more: https://aka.ms/ash-alerts

How can we make our incident communications more useful?

You can rate this PIR and provide any feedback using our quick 3-question survey: https://aka.ms/AzPIR/VLB8-1Z0

Hello from Seattle.

English (UK)

Feedback Trademarks Privacy & Cookies Terms of use

Microsoft
© 2024 Microsoft